

MASTERCLASS SERIES 5

Grid Failure Mastery

Fixing Generator Rejection, Eliminating Voltage Transients, & Automating Hand-overs for
Resilient Off-Grid Systems.



COURSE ROADMAP & TIMINGS

0:00 - 2:30

The Dilemma

Analyzing grid failures, rapid relay clicking loops, and active engine rejection parameters.

2:30 - 11:00

Act I: Rejection

Understanding Total Harmonic Distortion, Waveforms, and wide acceptance margins.

11:00 - 17:30

Act II: Automation

Implementing 2-Wire Dry Contact control signaling for low-voltage autonomy.

17:30 - 25:00

Act III: Hand-over

Generating physical, laminated distribution board scripts to guarantee safety transitions.

ACT I

The Generator Rejection Trap

Why sophisticated hybrid inverters reject mechanical engine generators, and how to configure them to stop the cycle.



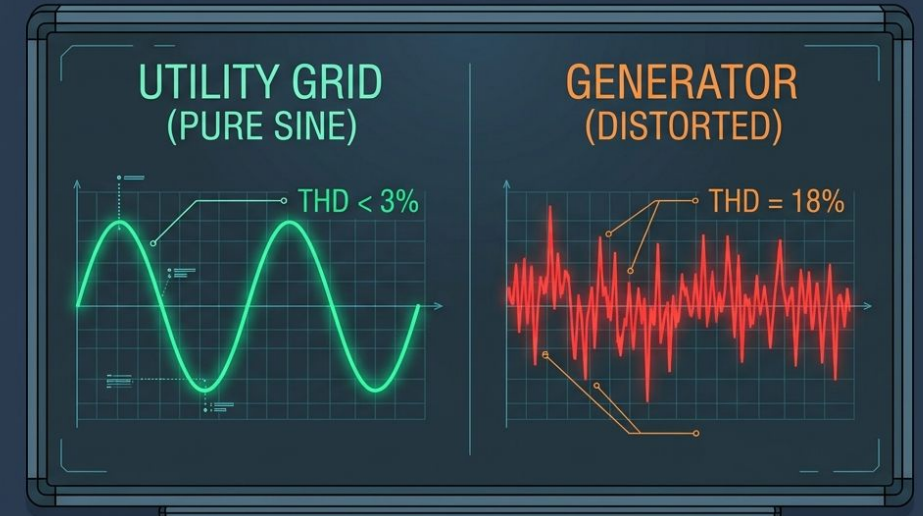
THE KVA SODA GLASS ANALOGY

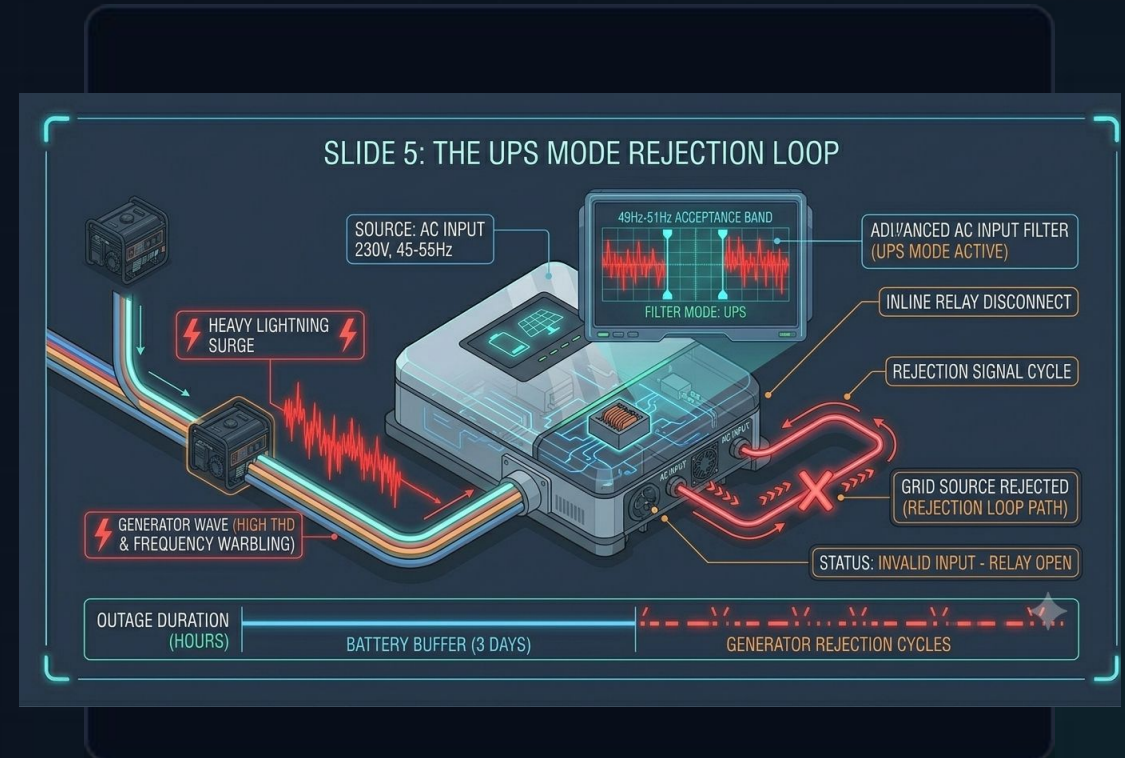
Visualizing Real vs. Reactive Power

To ensure you never get tricked by power metrics again, let's look at a helpful visual analogy: **A tall glass of carbonated soda.**

The liquid at the bottom of the glass represents your **True Power (kW)**—the actual working power that turns shafts, pumps water, and runs appliances.

The foam layer at the top represents **Reactive Power (kVAR)**. It does no real work but occupies critical volume inside the transmission glass (kVA).





TOTAL HARMONIC DISTORTION (THD)

Total Harmonic Distortion (THD) measures the electrical noise and deviation of an output waveform from a perfect sinus curve:

$$\text{THD (\%)} = \frac{\sqrt{\sum_{n=2}^{\infty} V_n^2}}{V_1} \times 100$$

Where V_1 is the fundamental frequency voltage and V_n represents the harmonic peak levels.

- ✔ **Inverter Target:** Stable systems operate with < 3% to 5% THD.
- ⚠ **Generator Profile:** Budget mechanical engines output 15% to 20% THD.
- ⊘ **Safety Trip:** Heavy inductive loads drag frequency down, prompting sensitive safety circuits to reject the connection.

RESOLVING THE REJECTION LOOP



UPS Mode (Default)

Enforces highly strict filtering. Rejecting incoming AC lines if frequencies shift beyond a tight **49 Hz - 51 Hz** envelope. Ideal for clean grids.



Generator / Wide Mode

Alters internal safety tolerances. Configures microprocessor limits to accommodate dynamic shifts from **45 Hz - 55 Hz** and voltage levels down to 170V.



Synchronized Charging

Modified parameters resolve repeating relay clicking. The inverter accepts input, locking step with mechanical output to charge battery banks safely.

ACT II

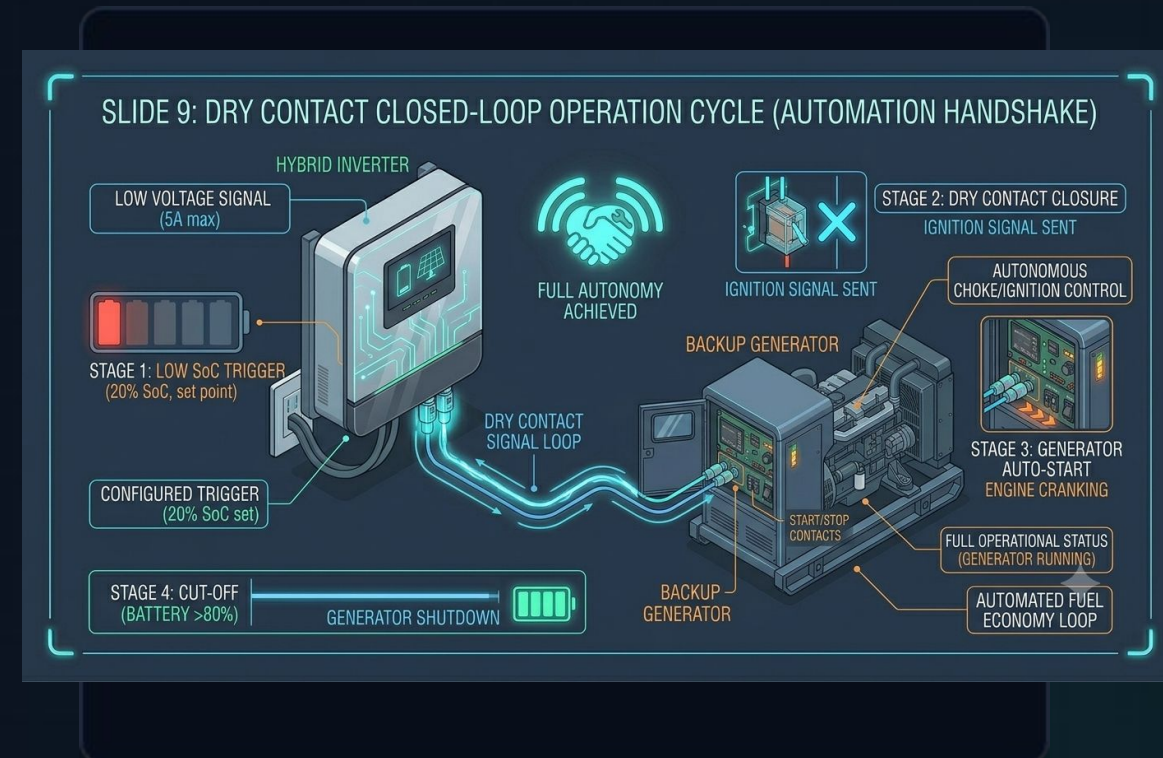
The Automated Clean Transition

Deploying resilient, hands-free hardware loops using dry contacts to fully automate backup start procedures.



2-WIRE DRY CONTACT CONTROL LOOP

- **Auxiliary Relays:** Dry contacts act as a physical finger pressing a start button automatically.
- 🛡️ **Safe Low-Voltage Control:** Signal loop runs low-voltage currents entirely separate from active home mains wiring.
- ⚙️ **Dynamic Integration:** Bridges battery state-of-charge parameters to electric engine starters.



THE 5-STEP AUTOMATION LOOP

0 1 Signal Line

Run simple 2-wire signaling line from inverter contacts directly to starter panel.

0 2 SoC Limit

Define low-battery limits on the inverter (e.g., 20% SoC trigger baseline).

0 3 Relay Close

Inverter brain closes contacts on low battery to complete the trigger loop.

0 4 Auto-Start

Mechanical generator choke opens, engine cranks, and automatically begins charging.

0 5 Cutoff Target

At 80% SoC, contacts open, breaking signal circuit to cut ignition spark safely.

ACT III

The Hand-Over Handshake

Shielding complex residential energy assets during real estate transfers with
standard printable operating scripts.



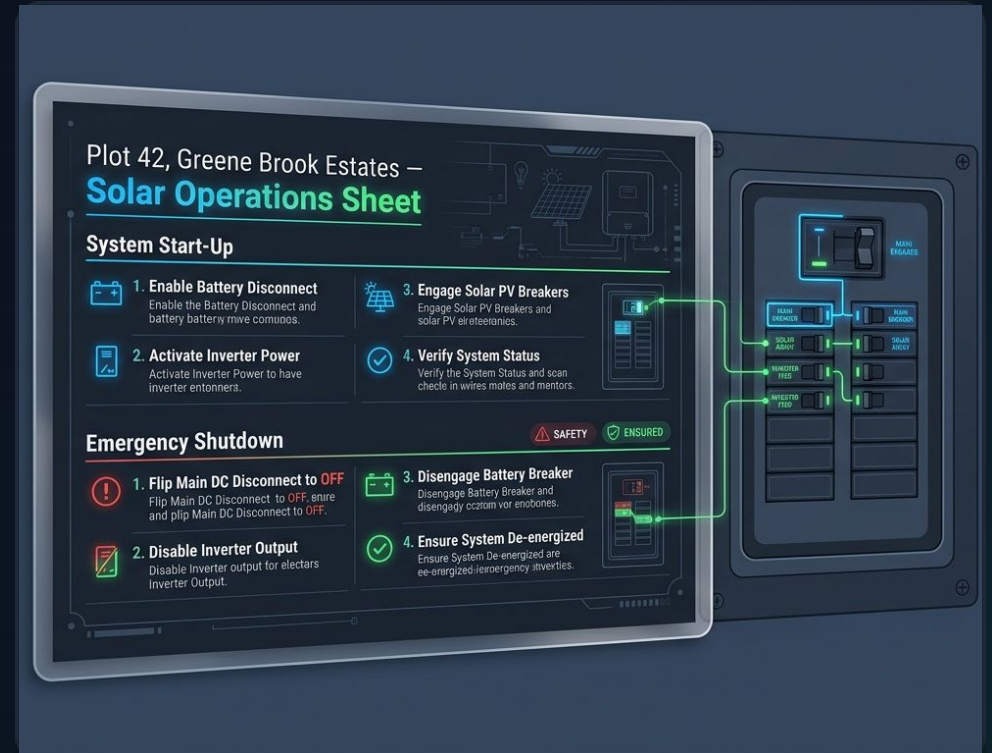
PROTECTING SOLAR REAL ESTATE ASSETS

Dynamic Operational Scripts

New occupants often don't understand battery depth-of-discharge parameters or inductive surge spikes.

Use the **Transfer Tab** on the Sizing Dashboard to generate plain-English operational scripts detailing limits, grounding nodes, and SPD targets.

Laminating and framing this script next to distribution boards protects complex off-grid hardware and boosts total transaction asset value.



SMART ENERGY WORKSPACE ACCESS



Signal Waves Tool: Toggle simulated wave structures on the live canvas to analyze regional harmonic noise.



Offline Reference Manual: Download standard wire diameter logs and installation guides under the Education tab.



Certified Hardware Center: Source certified hybrid inverters and Grade-A lifepo4 batteries with active warranty protection.



Active Interactive Workspace

Utilize calculations instantly at [index.html](#) to size system paths.

[Access our Telegram Engineer Portal](#)